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Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflameComprehensive kinetic modeling and experimental study of a fuel-rich, premixed *n*-heptane flameLars Seidel^b, Kai Moshhammer^{a,1}, Xiaoxiao Wang^b, Thomas Zeuch^c, Katharina Kohse-Höinghaus^{a,*}, Fabian Mauss^{b,*}^a Department of Chemistry, Bielefeld University, Universitätsstraße 25, D-33615 Bielefeld, Germany^b Thermodynamics and Thermal Process Engineering, Brandenburg University of Technology, Siemens-Halske-Ring 8, D-03046 Cottbus, Germany^c Institut für Physikalische Chemie, Georg-August-Universität Göttingen, Tammannstraße 6, D-37077 Göttingen, Germany

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ABSTRACT

An existing comprehensive kinetic hydrocarbon oxidation model has been augmented and revised for a detailed analysis of *n*-heptane flame chemistry. The analysis was enabled by experiments in which the detailed species composition in a fuel-rich flat premixed ($\phi = 1.69$) *n*-heptane flame at 40 mbar has been studied by flame-sampling molecular-beam mass spectrometry using electron impact ionization. Mole fraction profiles of more than 80 different species have been measured and compared against the new detailed kinetic model consisting of 349 species and 3686 elementary reactions. For all major products and most of the minor intermediates, a good agreement of the modeling results with the experimentally-observed mole fraction profiles has been found. The presence of low- and intermediate-temperature chemistry close to the burner surface was consistently observed in the experiment and the simulation. With the same kinetic model, *n*-heptane auto-ignition timing, flame speeds and species composition in a jet-stirred reactor have been successfully simulated for a broad range of temperatures (500–2000 K) and pressures (1–40 bar). The comprehensive nature and wide applicability of the new model were further demonstrated by the examination of various target experiments for other C_1 to C_7 fuels.

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1. Introduction

World's transportation is still driven by liquid fuels. Despite efforts to promote electric vehicles, this status is not likely to be changed in the near or mid-term future. A growing percentage of biofuels in blends from crude-oil-based fuels is one approach to reduce the fossil fuel dependence. However, this may lead to emission of different types of biofuel-specific hazardous species [1] such as aldehydes. To understand and model their formation pathways, it is necessary to understand the formation of pollutants in pure hydrocarbon fuels first.

n-Heptane is a major component in reference and surrogate fuels for internal combustion engines. The primary reference fuel (PRF) for octane rating is defined as a blend of *n*-heptane and *iso*-octane. Pure *n*-heptane is the simplest surrogate for diesel fuels. The surrogate ETRF (ethanol toluene reference fuel) has been

proposed as the future reference fuel for ethanol-containing gasoline fuels with a significant amount of *n*-heptane. Moreover, the degradation pathways for larger *n*-alkanes are analogous to *n*-heptane degradation. For these reasons, a vast interest arose in the combustion community to explore the combustion chemistry of *n*-heptane, which has become one of the most extensively studied hydrocarbons. However, even for *n*-heptane, the formation pathways of hazardous aldehyde, polycyclic aromatic hydrocarbon (PAH) or particulate emissions are still not fully understood and further research is needed that concentrates on these details.

One approach to unravel *n*-heptane combustion chemistry is to study the full species set as a function of reaction time in burner-stabilized flames. The first detailed study in this direction was reported in 1997 by Douté et al. [2]. The flame experiment was conducted at 60 mbar and equivalence ratios of $\phi = 0.7$ –2.0 (diluted by argon); molecular-beam-mass spectrometry (MBMS) was used as the analytical method. This work was later extended to a fuel-rich flame ($\phi = 1.9$ with N_2 dilution) at atmospheric pressure by the same group [3], while Ingemarsson et al. [4] investigated a stoichiometric flame (in air) at atmospheric pressure. Recently, stoichiometric low-pressure flames were investigated using molecular-beam mass spectrometry with synchrotron

* Corresponding authors.

E-mail addresses: kkh@uni-bielefeld.de (K. Kohse-Höinghaus), Fabian.Mauss@tdtvt.de (F. Mauss).¹ Current address: Combustion Research Facility, Sandia National Laboratories, Livermore, CA 94551, USA.

radiation for photoionization (PI-MBMS) [5–7], in these studies, the mole fraction profiles including many oxygenated species were reported. The influence of blending *n*-heptane with oxygenated fuels in premixed low-pressure flames ($\phi = 1.6$) was studied by Chen et al. [8].

In the experimental part of the present study, quantitative species measurements were conducted in a fuel-rich, premixed burner-stabilized low-pressure flame, in which significantly more species were quantified as in similar studies before [2–8]. A special emphasis was placed on the measurement of mole fraction profiles of important intermediates of PAH and soot formation pathways. Similarly, mole fraction profiles of many oxygenated intermediates were recorded that provide critical information for kinetic model development, also with regard to the oxidation of partially oxygenated compounds during biofuel combustion.

Due to its early definition as a reference fuel component in the 1930s for gasoline, *n*-heptane was and is an important target also in the field of combustion chemistry modeling. The first reaction scheme used for kinetic simulations was published in 1979 by Coats and Williams [9], followed by the work of Westbrook et al. [10,11] and Chakir et al. [12]. The first rule-based and automatically generated reaction mechanisms by Chevalier et al. [13] was also directed toward *n*-heptane modeling and included high- and low-temperature oxidation. This work was followed in the 1990s by Ranzi et al. [14] and Côme et al. [15]. Their models were examined against experiments in shock tubes and jet-stirred reactors. Later, Curran and co-workers published an extensively validated model for *n*-heptane oxidation [16], introducing 25 major classes of elementary reactions. Recent publications focused on the development of mechanisms comprising multiple fuels [17–19] or fuel blends [20,21]. The latest published kinetic model by Pelucchi et al. [22] focuses introducing additional reaction classes for the low-temperature chemistry by introducing additional reactions for hydroperoxide species. However, in all previous studies, the detailed modeling of species profiles in *n*-heptane flames has not been combined with a broad validation of the provided mechanisms against shock tube and jet-stirred reactor experiments that cover low-temperature oxidation chemistry; other fuels than *n*-heptane are not normally included in this validation either. Moreover, the modeling of *n*-heptane flame speeds proved to be difficult in early studies [23].

In 2007 some of the present authors reported the first detailed kinetic model for *n*-heptane oxidation, which was validated for the full range of combustors, pressures, fuel/oxidizer ratios and temperatures (500–2000 K) including shock tube experiments, jet-stirred and flow reactor measurements, ignition timing in a homogeneous charge compression ignition engine (HCCI), burner-stabilized flames and flame speed measurements. The widespread applicability of this reaction mechanism was demonstrated later in several studies that explored combustion processes beyond the initial target range [18,24–26] and for simulations of internal combustion engines [27,28]. This reaction mechanism [18] is the starting point for the modeling part of the present work. The detailed species measurements provided in this study enable a significantly deeper validation of the *n*-heptane flame chemistry compared to the original work, now including oxygenated species. This analysis was further supported by recent collaborative studies on the flame chemistry of butane and butene fuels [29,30] in which the C_4 chemistry of the original mechanism was significantly augmented. In this work the low-temperature *n*-heptane oxidation chemistry of the kinetic model has been critically inspected and revised.

Consequently, the reaction mechanism derived here provides a significant step beyond just another *n*-heptane flame chemistry model, validated against some newly measured flame data. The kinetic model augmented and presented in this work is a

comprehensive C_1 – C_7 hydrocarbon oxidation mechanism and its range of validation targets comprises a large number of fuels and fuel mixtures including methane, ethane, ethene, acetylene, propane, propene, acetylene/propene, *n*-butane, *iso*-butane, 1-butene, 2-butene, *iso*-butene, 1-hexene, *n*-heptane, toluene in flames and reactor experiments. This broad validation ensures that the mass flow analysis provides a realistic picture of *n*-heptane flame chemistry; especially with regard to pollutant formation (see discussion in [31]). The current hydrocarbon oxidation model, provided in the **Supplement Material**, consists of 349 species and 3686 reactions where forward and backward reactions are considered as individual reactions.

2. Experiment and data evaluation

2.1. Molecular-beam mass spectrometry (MBMS)

Experiments were carried out at Bielefeld University with a set-up described earlier in [30,32]. In brief, the instrument provides a two-stage Wiley-McLaren ion source equipped for electron ionization. It is coupled to a reflectron time-of-flight (TOF) mass spectrometer which enables a mass resolution of $m/\Delta m = 4000$. Thus, determination of the elemental composition of C/H/O species by their exact mass is possible in the relevant mass range (separation of oxygenated and non-oxygenated species is possible). The burner is mounted in an upright fashion in the center of a flame chamber (320 mm diameter, 550 mm height). The flame gases were sampled via a quartz nozzle with a 25° cone angle and a $\sim 500\ \mu\text{m}$ orifice diameter at the tip. A pressure reduction from 40 mbar in the flame chamber to $\sim 10^{-4}$ mbar in the 1st pumping stage interrupted further reactions and led to the formation of a molecular beam which passed through a copper skimmer (2 mm diameter) into the ionization chamber of the mass spectrometer at 10^{-6} mbar. Ionization relied on a pulsed electron beam of $\sim 10^9$ electrons/pulse with an energy distribution of 1 eV (*Full Width at Half Maximum*, FWHM) which permitted to detect all species in the same mass spectrum. Soft ionization using electron energies in the range between 10 and 18 eV were applied to minimize undesirable fragmentation processes. Measurements of intermediate species were performed at nominal energies of 10.5 eV, 12.0 eV, 13.5 eV, whereas an energy of 16.5 eV was applied for the determination of the major species, namely H_2 , H_2O , CO, O_2 , Ar, CO_2 , and *n*-heptane.

2.2. Flame and burner

A laminar flat premixed flame of *n*-heptane with a composition of *n*-heptane/oxygen/argon (10.0 mol%/65.0 mol%/25.0 mol%) was investigated at 40.0 mbar. The resulting stoichiometry is $\phi = 1.69$, and the respective cold gas velocity was 65.7 cm/s at 300 K, equivalent to a total mass flux of 8.312 g/min. The flow rates of oxygen and argon were regulated by calibrated mass-flow controllers (MKS, Aera) with a precision of $\pm 5\%$ since gas conversion factors were applied. The dosage of liquid *n*-heptane was realized by a syringe pump that injected the fuel into a heated vaporizer (130 °C). The entire Ar stream was fed through the vaporizer to ensure stable evaporation, thus an uncertainty of $\pm 5\%$ is also assumed for the *n*-heptane flow. In due consideration of these uncertainties, the stoichiometry of the flame covers a range from 1.61 to 1.78.

The flame was stabilized on a home-built flat-flame burner which features a sintered and water-cooled bronze matrix with a diameter of 64.0 mm. The burner was mounted on a translational stage that is movable in vertical and horizontal directions. By varying the distance to the nozzle, species mole fraction profiles were obtained along the centerline of the flame as a function of height

h above the burner. The pressure was kept constant during the measurements using a downstream butterfly valve (MKS) that regulated the pumping speed of a mechanical pump.

2.3. Data evaluation

The data quantification method was adapted from previous work [30,32]. A detailed description of the procedure used in the data evaluation is given in the supplement of [30]. Thus, only the main aspects of the evaluation methods are reported here.

The integrated ion signal S_i of a species i at the energy E is given by:

$$S_i(E) = x_i \cdot c \cdot \varphi \cdot SW \cdot D(M_i) \cdot FKT(h) \cdot \int \sigma_i(\tau) \cdot f(E - \tau) d\tau. \quad (1)$$

Here, x_i is the mole fraction of species i , c an instrument factor, φ the number of electrons, SW the number of sweeps, $D(M_i)$ the mass discrimination factor of species i . FKT is a temperature- and thus position-dependent sampling function, and $\int \sigma_i(\tau) \cdot f(E - \tau) d\tau$ is the convolution of the energy distribution of ionizing particles with the ionization cross section (see [30] for details), with τ as integration variable. Eq. (1) may be simplified if a species is measured relative to a reference species R :

$$\frac{S_i}{S_R} = \frac{x_i}{x_R} \cdot \frac{D(M_i)}{D(M_R)} \cdot \frac{\int \sigma_i(\tau) \cdot f(E - \tau) d\tau}{\int \sigma_R(\tau) \cdot f(E - \tau) d\tau} = \frac{x_i}{x_R} \cdot k_{i/R}(E). \quad (2)$$

Here, $k_{i/R}(E)$ is a calibration factor which combines all species-dependent parameters and which is related to the ionization energy. Because of the broad energy distribution of the electron beam (FWHM = 1 eV), the instrument enables a species overview from one single mass spectrum. Argon was thus used as a reference signal at all measured energies. The calibration factors $k_{i/Ar}(E)$ were generated from cold gas measurements of known gas compositions (*direct calibration*), or were estimated by the RICS (*relative ionization cross sections*) method [33], or by a convolution of literature ionization cross sections with the known kinetic energy distribution of ionizing electrons [34]. With the respective calibration factor known, the mole fraction of each species was obtained based on the argon mole fraction profile from the main species calculation (see below). Additional signal corrections were applied to correct for ^{13}C and ^{18}O isomers and background signals. To reduce the influence of fragmentation reactions on mole fraction profiles, signals were corrected for fragmentation coming from species with known fragmentation patterns which were obtained by cold-gas (calibration) energy scans. Fragmentation of the fuel was considered for all ionization energies and all possible C/H fragments based on independent cold-gas measurements. Moreover, fragmentation of intermediate species was considered for several species. In addition, the scan with the lowest possible energy was used for most of the species.

Mole fractions of major species including n -heptane, O_2 , H_2 , H_2O , CO , CO_2 , and Ar were calculated using Eq. (2). All signals were taken from the 16.5 eV scan. The determination of the necessary calibration factors was performed using an internal calibration strategy [30] based on the elemental C, H, and O balances and on a single cold-gas calibration measurement of a defined CO/CO_2 mixture. The argon mole fraction was calculated as the balance between the sum of all major species and unity. To account for early consumption of small amounts of O_2 and the fuel near the burner surface, the individual element balances were used to determine calibration factors for these species. Furthermore, a correction of the raw signals for their background (caused by accumulation of background gas in the ionization chamber) was performed for some species.

Although many corrections have been carefully considered, mole fractions may be affected by several uncertainties. Even if care is taken to correct for fragmentation (especially from the fuel), smaller contributions for species with unknown fragmentation patterns, e.g. radicals, cannot be excluded. Uncertainties in the mole fractions depend on the accuracy of the calibration factors and on the respective electron ionization cross sections. The accuracy is about 20–30% for directly calibrated species, but factors of 2–4 uncertainty are possible if cross sections must be estimated.

For the main species an accuracy of $\pm 15\%$ can be achieved by using the calibration method described above.

2.4. Temperature

Temperature profiles were determined and used as input parameters for the kinetic modeling, recognizing that MBMS is an invasive technique and that the flow and temperature fields of the laminar flame are perturbed by the sampling probe. Typically the perturbation influence on the measured species profiles is lowest for sufficiently large burner-to-cone distances [35], and the influence on the temperature field can be described adequately by using a perturbed temperature profile [36,37]. Perturbed temperature profiles were obtained here from the temperature dependence of the sampling rate. The gas flow rate $\dot{N}$ through the sampling orifice can be described by the following relation:

$$\dot{N} \propto p_{1st} \propto \frac{S_{Ar}}{x_{Ar} \cdot \varphi} = C \cdot \sqrt{\frac{\gamma}{M \cdot T}} \cdot \left(\frac{2}{\gamma + 1}\right)^Z \quad \text{with } Z = (\gamma + 1)/2(\gamma - 1). \quad (3)$$

Here, S_{Ar} is the integrated ion signal and x_{Ar} the mole fraction of argon, φ is the number of electrons, γ is the heat capacity coefficient (C_v/C_p), and $\bar{M}$ is the mean molar mass. C is an instrument-specific constant, which can be obtained at a defined position with known temperature T and known composition. Such a position can be most reliably identified in the exhaust gas region where the temperature can be determined without the need of high spatial resolution and the gas composition is well known.

The temperature of the calibration point in the exhaust gas was measured by laser-induced fluorescence (LIF) of seeded NO at a distance of $h = 16$ mm without the presence of the sampling cone. Small amounts of NO (<0.5%) were added to the fuel stream for this measurement. NO was excited in the A–X (0–0) band in the range of 225.047–225.124 nm with a dye laser (Coherent, Scan Mate Pro) which was pumped by an excimer laser (Coherent, CompEx150) at 50 Hz. The broadband LIF emission was detected with a CCD camera (Andor, iStar DH734), and the temperature was evaluated with an accuracy of $\pm 5\%$ by direct fitting of the spectra.

Temperatures for each height above the burner can then be obtained from the recorded pressure in the 1st stage (measured by a pressure sensor) or from the height-dependent argon signal and its calculated mole fraction profile.

3. Chemical model

An existing chemical kinetic model was improved to capture and describe the combustion of n -heptane in more detail than before. It is based on a series of former development steps which include a broadly validated reaction mechanism for the C_1 – C_4 chemistry [31], the combustion of toluene [38] as well as of the butane [29] and butene [30] isomers. The hierarchically constructed mechanism was validated against low-pressure flame speciation data measured in this study, laminar flame speeds at different pressures as well as ignition delay time measurements [39–41] and species composition results from jet-stirred reactors [42]. Detailed information about the literature sources of kinetic

and thermodynamic properties for most of the reactions and species of the present model is provided in extensive earlier works [18,31]. In these studies the overall mechanism compilation strategy, summarized in the introduction, is explained in more detail. Changes of these data, introduced in this work, are discussed below. We note that in early studies and in the present work estimates of the kinetic data for reactions without literature sources were in some cases based on analogy considerations with reactions where data is available.

The C₁–C₄ chemistry (in the following named “base chemistry”) was updated for a few chemical steps. In the older versions, the pressure dependence of the C–H fission rate in the vinyl radical (C₂H₃) was interpolated using kinetic data for several pressures provided by Warnatz and co-workers [43]. In the present work we formulate the pressure-dependent kinetics of the C₂H₃(+M) = H + C₂H₂(+M) reaction by the Troe ansatz, using the recommendation of Baulch et al. [44]. Similarly, the termolecular reaction CH₃ + H(+M) = CH₄(+M) now follows the suggestion of the CEC Group on Evaluation of Kinetic Data for Combustion Modeling [45]. Here, the Arrhenius parameters for the collision frequency of $k(0)$ and $k(\infty)$ were reduced by 40%, respecting the uncertainties in the CEC Group suggestions [45]. These changes were implemented in order to simultaneously predict laminar flame speeds for a set of smaller fuels (methane, ethane, ethene, propane, propene) and *n*-heptane in agreement with recent measurements [46,47] at 1 bar and at elevated pressures [48]. In addition we included a high-temperature sub-mechanism for *n*-pentane and *iso*-pentane to inspect the formation pathways of both species in the *n*-heptane flame. The sub-mechanisms were compiled using the same rules as for the *n*-heptane oxidation scheme. Further a decomposition pathway for dimethyl ether was added based on the suggestion of Fischer et al. [49].

Also the chemistry leading to polycyclic aromatic hydrocarbon formation was modified. The formation of the important cyclopentadiene (cyc-C₅H₆) from smaller alkanes is taken into account by two reactions: C₂H₂ + C₃H₅ = cyc-C₅H₆ + H [50] and C₃H₄ + C₃H₅ = cyc-C₅H₆ + CH₃ [51]. The effect of these changes will be discussed in Section 4.

The pathways for benzene formation via fulvene were discussed in detail in a recent study on 1-hexene flame chemistry [52]. In that work the fulvene chemistry has been included and the thermodynamic data of several species including C₃H₃ (propargyl), C₃H₄ (allene), C₃H₄P (propyne), C₃H₅ (allyl), C₃H₆, *n*-C₃H₇, and *i*-C₃H₇ have been updated. In the present work the soot precursor chemistry has been further augmented by including build-up reactions for larger polycyclic aromatic species such as naphthalene and 3-ring species. Their kinetic data is based on [53] and was extended by reactions from Blanquart et al. [54], D’Anna et al. [55], Raj et al. [56], Lindstedt et al. [57], Marinov et al. [58], Harding et al. [59] and Zhang et al. [60].

The sub-mechanism for *n*-heptane oxidation is largely similar to that published by some of the present authors in 2007 [18], which is based on rules described by Curran et al. [16]. However, the detailed information on species composition in a fuel-rich flame, provided in the present and earlier works [29,30], was used for refining the oxidation scheme in several key points. Firstly, we can consider now more possible decomposition products due to the extended base chemistry. H-atom abstraction reactions from the fuel by highly oxygenated methane derivatives as suggested by Curran et al. [16] were removed. With this change the main H-atom abstraction route at low temperatures is found to be dominated by the hydroxyl radical (OH). For this reaction type we follow the suggested kinetic data of Sivaramakrishnan and Michael [61]. Using their data improves model predictions in the low temperature range compared to those reported in earlier studies [62,63]. The addition of radical species to alkenes was

implemented in accordance with the previous work on the flame chemistry of butane isomers [30]. In contrast to this earlier study we now treat the H-atom abstraction from formed alkenes similar to the H-atom abstraction from the fuel molecule itself. The decomposition of alkenyl radicals is represented in more detail as suggested by Mehl et al. [64]. The revised kinetic data for the thermal decomposition reactions follow the study of El Bakali et al. [3], who suggested rates from theoretical predictions [65,66]. The β-C–C-fission for secondary–secondary C-atom bonds is favored over secondary–primary C-atom bonds, which is in agreement with recently published kinetic models [21]. We note that the influence of retro-ene type reactions (see discussion in [67]) of alkenes have been tested for all experimental targets. Only a marginal impact on simulation results was found and therefore this reaction path has not been included. The mechanism files can be found in the Supplement Material.

4. Results and discussion

The discussion of the experimental data and modeling results is organized as follows. First, the disturbance of the flame by the sampling cone and its effect on flame temperature is discussed. This is motivated by its importance regarding the comparison of model predictions and experimental data. Thereafter, based on a C-atom mass flow analysis, the main degradation pathways of *n*-heptane under fuel-rich flame conditions are discussed. The validity of this analysis is corroborated by a detailed comparison of experimental data and kinetic modeling results for a number of species. An overview of the experimental and simulated peak mole fractions as well as their heights above the burner are given in Table 1. We first present main product formation, followed by production and consumption of important intermediate species. Finally, pathways leading to soot precursors and the formation of oxygenated species are analyzed. In the last section the model predictions are compared to shock tube and jet-stirred reactor experiments as well as *n*-heptane flame speed measurements including unburned gas temperature and pressure variation. All calculation were performed with the current version of the LOGEsoft package [68]. Details about the numerical methods applied for solving the differential equations of the different reactor models can be found in the manual of the LOGEsoft package.

4.1. Speciation in a fuel rich *n*-heptane flame

4.1.1. Temperature profile

For the modeling calculation of the burner-stabilized flame, mixture average and thermal diffusion were considered. The measured disturbed temperature profile was used as input parameter as described in Section 2.4. The influence of the probe disturbance was estimated by comparing the disturbed profile to the computed temperature profile in Fig. 1. The temperature profile is calculated from the energy conservation equation, which considers the heat flux to the burner. For all stable species we calculate the concentrations on the burner surface from the gas composition in the feed, and the species diffusion close to the boundary. For all major radicals we consider recombination at the burner surface. The undisturbed flame stabilizes above the burner through heat loss, which is calculated from the temperature gradient at the burner surface. The heat loss determines the maximum flame temperature, which is below the adiabatic flame temperature. From flame theory it is known that the flame speed is proportional to the maximum flame temperature. Hence, the heat loss of the stable flame corresponds to a maximum flame temperature for which the burner exit velocity and the reduced flame velocity are equal. The close agreement between predicted and measured maximum flame temperature

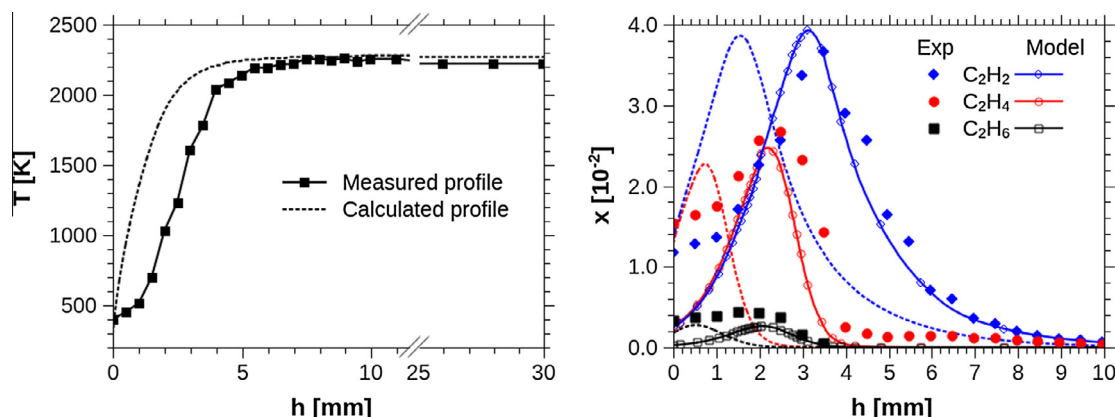


Fig. 1. Left: Measured (disturbed) and calculated temperature profiles. Right: Experimental profiles for C_2H_2 , C_2H_4 , and C_2H_6 (filled symbols) and predictions using the simulated (solid lines with open symbols) and the disturbed temperature profiles (dotted lines).

gives us confidence in both, the measured temperature profile and the predicted laminar flame velocity.

The strongest difference could be observed close to the burner surface. A shift in the onset of the temperature profile by 2 mm is found which corresponds well to the shift in the predicted species concentration profiles. The profiles of C_2H_2 , C_2H_4 , and C_2H_6 for both temperature profiles are shown in the right panel of Fig. 1. It can be seen that the predicted global peak concentrations differ by less than 10% and that the profiles are shifted in the same direction with the same distance as the temperature profile. This underlines the hypothesis that the probe mainly influences the data close to the burner. The peak temperatures differ by about 5 K. In the experiment the temperature decreases due to radiation losses, with a small impact on the prediction of the mole fraction profiles. A further important observation is that the simulation with calculated energy conservation can predict an O_2 concentration at the burner surface ($h = 0$ mm) close to the measured value. Figure 2 illustrates the very good agreement of calculated and measured stable species concentrations under equilibrium conditions at $h > 6.0$ mm, further underlining the quality of the measured temperature. Rate kinetic data cannot influence the species concentrations at these heights and the thermodynamic data of these species are well known. We conclude that the probe distorts measurements mainly at the first two millimeters above the burner, by shifting all species profiles. Thereafter very little effect can be reported. The probe effect can be taken into account by

using the measured temperature profile as it was proposed before. A recent numerical study analyzes such probe-sampling effects [69]. In the following the measured, disturbed, temperature profile was used as described in Section 2.4.

4.1.2. *n*-Heptane consumption

The C-atom mass flow analysis is schematically illustrated in Fig. 3. Due to the complex flame chemistry, only the main degradation pathways are shown. The detailed mass flow analysis is presented in the Supplement Material, in which the integrated mass flows over the whole flame are given (Figs. S5 and S9). The consumption of *n*-heptane starts with H-atom abstraction by H-atoms which are present with significant concentrations close to the burner surface caused by (thermo)-diffusion. This channel accounts for about 95% of the fuel consumption. The fuel radicals then react further via three different pathways:

- (1) β -C–C scission,
- (2) β -C–H scission producing alkenes, and
- (3) low-temperature oxidation with early formation of oxygenated C_7 intermediates.

The relevance of the low-temperature pathway in flames is surprising and has not been reported before. The flow analysis shows that the predominant pathway is the fast decomposition of the C_7H_{15} radicals by β -C–C scission contributing to their consumption with more than 80%. This decomposition pathway was much less pronounced in simulations with the previous kinetic model, in which slower rate coefficients for the C_7H_{15} decomposition by β -C–C fission were applied [18]. The direct formation of C_7H_{14} alkenes by β -C–H fission contributes with about 2.5% to the consumption of the fuel radical under the present conditions. In contrast to the previous work of some of the authors, this route is not the main pathway anymore. The changes in the present kinetic model, responsible for the altered mass flows, were motivated by a disagreement between measured and calculated C_7H_{14} mole fractions. We will return to this point in the Section 4.1.4.

We first discuss the unexpected presence of low-temperature *n*-heptane oxidation chemistry close to the burner surface. The low-temperature chemistry pathway is initiated by the addition of O_2 to the fuel radicals and involves the characteristic subsequent chemical steps of internal H atom abstraction and the formation of cyclic ethers by isomerization and OH loss of the intermediate hydroperoxide radical. This reaction sequence was observed for all four C_7H_{15} isomers and contributes between 1.9% (C_7H_{15} -R1) and 6.5% (C_7H_{15} -R4) to the consumption of the fuel radical. Due

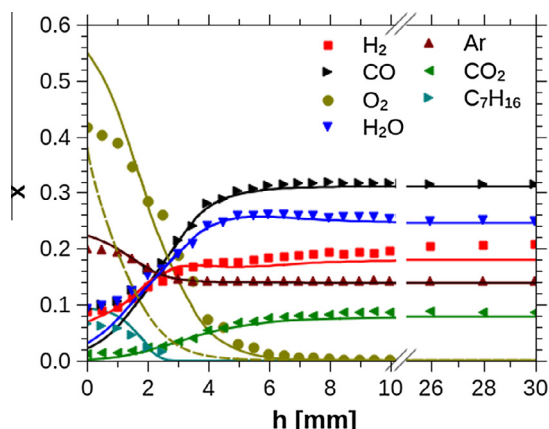


Fig. 2. Mole fractions of major products. Symbols represent measurements; solid lines: model predictions using the measured temperature profile; dotted line: O_2 prediction from the energy conservation (without temperature profile).

Table 1
Measured and simulated species concentrations.

Species	<i>E</i> (eV)	IP (eV)	Calculated as	Calibration method	σ	$x_{\max}$ (Exp.)	<i>h</i> (mm)	$x_{\max}$ (Sim.)	<i>h</i> (mm)	Exp/Sim
H ₂	16.5	15.426	Hydrogen	See text	–	2.08E–01	30.0	1.80E–01	29.96	1.15
CH ₂	10.5	10.396	Methylene radical	RICS (CH ₄)	[74]	6.14E–05	2.5	8.38E–04	4.01	13.65
CH ₃	10.5	9.84	Methyl radical	RICS (CH ₄)	[74]	5.14E–03	2.5	1.26E–02	2.92	2.45
CH ₄	12	12.61	Methane	Direct	[75]	6.96E–03	3.0	1.18E–02	2.53	1.70
C ₂ H	12	11.61	Ethynyl radical	Convolution	RICS (C ₂ H ₂)	4.95E–05	3.0	8.78E–05	2.96	1.77
C ₂ H ₂	12	11.4	Acetylene	Direct	[76]	3.67E–02	3.5	3.88E–02	3.12	1.06
C ₂ H ₃	10.5	8.25	Vinyl radical	RICS (C ₂ H ₄)	[77]	4.32E–05	2.0	7.17E–04	2.93	16.59
C ₂ H ₄	12	10.514	Ethene	Direct	[78]	2.67E–02	2.5	2.44E–02	2.22	0.91
HCO	10.5	8.12	Formyl radical	Convolution	[77]	4.08E–05	2.0	9.81E–05	2.64	2.40
C ₂ H ₅	10.5	8.117	Ethyl radical	RICS (C ₂ H ₆)	[76]	3.36E–03	2.0	9.53E–04	2.25	0.28
CH ₂ O	12	10.88	Formaldehyde	Convolution	[77]	4.02E–03	2.0	3.81E–03	2.28	0.95
C ₂ H ₆	12	11.52	Ethane	Direct	[78]	4.44E–03	1.5	2.33E–03	2.11	0.53
CH ₃ O	12	10.72	Methoxy radical	RICS (CH ₃ OH)	[76]	1.74E–04	0.0	5.39E–05	1.57	0.31
CH ₃ OH	12	10.84	Methanol	Direct	[79]	5.60E–04	0.5	2.59E–04	1.50	0.46
Propadienylidene	12	10.43	Propadienylidene	Convolution	RICS (Propene)	5.10E–06	2.0	5.50E–05	4.16	10.77
C ₃ H ₃	10.5	8.67	Propargyl radical	RICS (C ₃ H ₆)	[77]	3.11E–04	3.0	8.31E–04	3.48	2.67
C ₃ H ₄	12	10.36	Propyne	RICS (C ₃ H ₆)	[77]	4.37E–03	3.0	1.99E–03	2.95	0.46
C ₃ H ₅	10.5	8.13	Allyl radical	RICS (C ₃ H ₆)	[77]	8.69E–04	2.5	2.25E–03	2.81	2.59
C ₂ H ₂ O	12	9.617	Ketene	Convolution	RICS (C ₂ H ₂ O)	5.95E–04	3.0	4.09E–04	2.44	0.69
C ₃ H ₆	12	9.73	Propene	Direct	[78]	4.44E–03	2.5	1.39E–02	2.28	3.13
C ₂ H ₃ O	12	10.85	Vinyloxy radical	Convolution	RICS (C ₃ H ₆ O)	1.52E–04	0.5	3.12E–04	2.37	2.06
C ₃ H ₇	10.5	8.09	<i>n</i> -Propyl radical	RICS (C ₃ H ₈)	[76]	1.99E–04	1.5	1.60E–04	1.77	0.81
C ₂ H ₄ O	12	10.229	Acetaldehyde	Direct	–	8.16E–04	2.0	9.59E–04	2.02	1.18
C ₃ H ₈	12	10.94	Propane	Direct	[76]	2.52E–03	2.0	6.72E–03	2.14	2.67
C ₂ H ₅ O	10.5	9.11	Ethoxy radical	RICS (C ₂ H ₅ OH)	[76]	1.56E–05	0.0	–	–	0.00
CH ₂ O ₂	12	11.3	Formic acid	Convolution	RICS (CO ₂)	7.05E–05	1.0	–	–	0.00
C ₂ H ₆ O	12	10.48	Ethanol	Direct	–	1.03E–03	0.5	4.56E–04	1.89	0.44
C ₄ H ₂	12	10.17	1,3-Butadiyne	Convolution	[77]	1.33E–03	3.5	1.57E–03	3.46	1.18
C ₄ H ₃	10.5	8.06	<i>i</i> -C ₄ H ₃ radical	Convolution	RICS (C ₄ H ₈)	1.25E–06	3.0	4.62E–05	2.55	37.08
C ₄ H ₄	10.5	9.58	1-Buten-3-yne	Convolution	[77]	5.40E–04	3.0	5.79E–04	3.04	1.07
C ₄ H ₅	10.5	7.95	But-2-yn-1-yl radical	RICS (C ₄ H ₈)	[76]	1.02E–05	2.5	2.64E–05	3.08	2.60
C ₄ H ₆	10.5	9.072	1,3-Butadiene	Convolution	[77]	1.65E–03	2.5	2.13E–03	2.61	1.29
C ₃ H ₃ O	12	7	1-Oxoprop-2-enyl	RICS (C ₃ H ₆ O)	[76]	6.77E–06	1.5	4.65E–06	1.33	0.69
C ₄ H ₇	10.5	8.04	But-3-en-1-yl radical	RICS (C ₄ H ₈)	[76]	8.24E–05	2.0	1.08E–04	2.33	1.32
C ₃ H ₄ O	12	10.11	2-Propenal	RICS (C ₃ H ₆ O)	[76]	2.93E–04	2.0	3.32E–05	2.81	0.11
C ₄ H ₈	10.5	9.55	1-Butene	Direct	[77]	5.68E–03	2.0	3.69E–03	2.04	0.65
C ₄ H ₉	10.5	8.02	1-Butyl radical	RICS (C ₄ H ₈)	[76]	1.16E–05	1.5	4.76E–06	1.70	0.41
C ₂ H ₂ O ₂	10.5	10.2	Glyoxal	Convolution	[76]	1.89E–05	1.0	–	–	0.00
C ₃ H ₆ O	10.5	9.703	Acetone	Direct	[76]	1.67E–04	1.5	5.39E–04	1.85	3.23
C ₄ H ₁₀	12	10.53	<i>n</i> -Butane	Direct	–	5.70E–04	1.5	6.17E–04	1.95	1.08
C ₃ H ₇ O	10.5	9.2	1-Methyl ethoxy radical	Convolution	RICS (C ₄ H ₈ O)	1.96E–06	0.5	2.25E–07	0.86	0.11
C ₂ H ₄ O ₂	12	10.65	Acetic acid	Convolution	[80]	3.66E–05	4.0	–	–	0.00
C ₃ H ₈ O	10.5	10.22	1-Propanol	Convolution	[76]	2.21E–05	1.5	–	–	0.00
C ₅ H ₃	10.5	8.28	CH ₂ CCCCH	RICS (C ₅ H ₈)	[76]	1.48E–06	3.0	9.08E–06	3.93	6.13
C ₅ H ₄	10.5	9.5	1,3-Pentadiyne	RICS (C ₅ H ₈)	[76]	9.07E–05	3.0	–	–	0.00
C ₅ H ₅	10.5	8.41	Cyclopentadienyl radical	RICS (C ₅ H ₈)	[76]	1.22E–05	3.0	3.60E–05	3.04	2.95
C ₅ H ₆	10.5	8.57	1,3-Cyclopentadiene	RICS (C ₅ H ₈)	[76]	9.93E–05	2.5	5.82E–05	2.57	0.59
C ₅ H ₇	10.5	7.25	1,4-Pentadien-3-yl radical	RICS (C ₅ H ₈)	[76]	3.34E–06	2.5	5.34E–09	2.37	0.00
C ₄ H ₄ O	10.5	10.28	2-Butynal	Convolution	RICS (C ₄ H ₈ O)	6.29E–05	2.5	–	–	0.00
C ₅ H ₈	10.5	8.86	Isoprene	Direct	[76]	1.61E–04	2.5	5.99E–04	2.39	3.73
C ₅ H ₉	10.5	7.3	1-Penten-3-yl radical	RICS (C ₅ H ₁₀)	[76]	4.89E–06	1.5	6.37E–07	1.58	0.13
C ₄ H ₆ O	10.5	9.75	2-Butenal	Convolution	RICS (C ₄ H ₈ O)	5.39E–05	0.0	5.55E–06	2.79	0.10
C ₅ H ₁₀	10.5	9.49	1-Pentene	Direct	[76]	1.57E–03	2.0	1.66E–03	2.11	1.06
C ₅ H ₁₁	10.5	7.86	3-Pentyl radical	RICS (C ₅ H ₁₂)	[76]	1.29E–04	1.0	1.18E–05	1.76	0.09
C ₄ H ₈ O	10.5	9.82	Butanal	Convolution	RICS (C ₄ H ₁₀ O)	4.96E–05	1.5	1.11E–08	1.71	0.00
C ₅ H ₁₂	12	10.28	Pentane	Direct	[81]	2.48E–04	2.0	4.47E–04	1.74	1.80
C ₄ H ₉ O	12	9.22	<i>n</i> -Butoxy radical	Convolution	RICS (C ₄ H ₁₀ O)	9.35E–06	1.0	5.11E–10	0.94	0.00
C ₆ H ₂	10.5	9.5	1,3,5-Hexatriyne	Convolution	[77]	3.59E–05	3.5	4.00E–05	3.62	1.11
C ₃ H ₆ O ₂	12	10.44	Propionic acid	Convolution	RICS (C ₂ H ₆ O)	3.22E–06	3.5	–	–	0.00
C ₄ H ₁₀ O	12	9.99	1-Butanol	Convolution	[79]	2.58E–06	1.5	1.79E–07	1.57	0.07
C ₆ H ₄	10.5	9.03	Benzyne	Convolution	RICS (C ₆ H ₂)	1.23E–05	3.5	8.96E–07	4.12	0.07
C ₆ H ₅	12	8.32	Phenyl radical	Convolution	RICS (C ₆ H ₆)	3.56E–07	3.0	6.46E–09	3.15	0.02
C ₆ H ₆	10.5	9.24	Benzene	Direct	[77]	7.87E–05	3.0	1.06E–04	2.72	1.35
C ₆ H ₉	10.5	7.54	3-Cyclohexenyl radical	RICS (C ₆ H ₁₀)	[76]	7.54E–07	2.0	2.47E–07	2.30	0.33
C ₆ H ₁₀	10.5	8.95	Cyclohexen	Direct	–	6.04E–05	2.0	2.50E–06	2.33	0.04
C ₆ H ₁₂	10.5	9.88	Cyclohexane	Direct	–	9.91E–04	2.0	5.99E–04	2.08	0.61
C ₇ H ₈	10.5	8.828	Toluene	Direct	[76]	8.59E–06	3.0	7.30E–06	2.78	0.85
C ₆ H ₆ O	12	8.49	Phenol	Convolution	RICS (C ₆ H ₆)	3.90E–06	3.0	6.33E–06	2.63	1.62
C ₇ H ₁₄	10.5	9.34	1-Heptene	Convolution	RICS (C ₇ H ₁₆)	2.88E–04	1.5	1.09E–03	1.72	3.78
C ₇ H ₁₅	10.5	7.9	Heptyl radical	Convolution	RICS (C ₇ H ₁₆)	7.43E–06	1.5	1.23E–04	1.69	16.51
C ₇ H ₁₆	16.5	0	<i>n</i> -Heptane	Direct	[82]	6.61E–02	0.0	–	–	0.00
C ₈ H ₈	10.5	8.464	Styrene	RICS (C ₇ H ₈)	[76]	1.31E–06	3.0	–	–	0.00
C ₈ H ₁₀	10.5	8.8	Ethylbenzene	RICS (C ₇ H ₈)	[76]	5.35E–06	3.0	–	–	0.00
C ₉ H ₈	10.5	8.14	Indene	Convolution	RICS (C ₆ H ₆)	4.56E–07	3.0	5.17E–08	2.81	0.11
C ₁₀ H ₈	10.5	8.144	Naphthalene	Convolution	RICS (C ₆ H ₆)	2.07E–07	3.0	1.10E–07	2.93	0.53

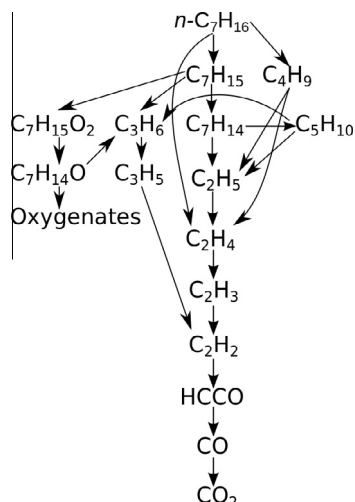


Fig. 3. Schematic of the main carbon atom flow in the fuel-rich *n*-heptane flame. The paths of several carbon fragments are neglected and isomers are not considered.

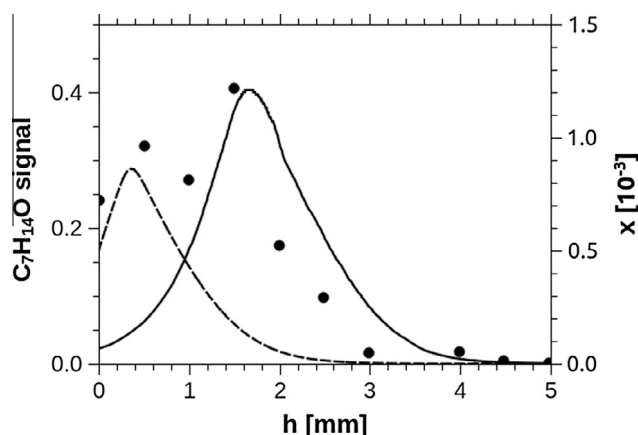


Fig. 4. Comparison of $C_7H_{14}O$ signal (symbols, left axis) and model prediction (right axis) using the measured temperature profile (solid line) and the model prediction without temperature profile (dotted line).

to its complexity the present discussion focuses exemplarily on the primary C_7H_{15} -R1 radical. For this radical, the low-temperature pathway is only active up to $h = 2.5$ mm with its maximum flux peaking at 1.5 mm (using the experimental temperature profile in the simulation). The competing unimolecular decomposition producing C_7H_{14} -D1 starts at about $h = 2$ mm and peaks at 2.5 mm. This is in line with the general assumptions about the transition from low- to high-temperature combustion chemistry. At a height of 2 mm the experimental temperature reaches a value of 1000 K which is the typically assumed temperature above which the high-temperature combustion chemistry starts. This analysis indicates that a part of the reaction zone lies in the intermediate-temperature regime. The LOGEsoft kinetic tool facilitates a zone-specific mass flow analysis, which has been performed between 0.8 and 2.0 mm above the burner, showing the significant production of peroxy radicals. A figure illustrating this result (Fig. S10) is provided in the Supplement Material along with a sensitivity analysis (Fig. S11). The same low-temperature chemistry was found when using no temperature profile for the simulation with a flux peaking at about $h = 0.4$ mm. It was not possible to unambiguously identify oxygenates with the molecular formula $C_7H_{15}O_2$ in the flame, but a clear $C_7H_{14}O$ signal was detected, as shown in Fig. 4.

Both, the species signal and the model prediction exhibit a peak near 1.7 mm where a temperature of about 800 K was measured for the disturbed temperature profile. In the absence of reference data, the $C_7H_{14}O$ signal has not been calibrated. In the model, $C_7H_{14}O$ is formed via the reaction sequence $n-C_7H_{16} \rightarrow C_7H_{15}O_2 \rightarrow$ -alkyl peroxy radical isomerization $\rightarrow$ formation of the cyclic ether $C_7H_{14}O$. This pathway is typical for *n*-alkane oxidation in the low-temperature regime and is commonly observed in auto-ignition chemistry at similar temperatures. Consequently, the degradation of cyclic ethers (these are the only species with the molecular formula $C_7H_{14}O$ in the kinetic model) is captured by the C_1 – C_7 chemistry of the model, which is validated for low- and intermediate-temperature experiments ([18] and Section 4.2). A species with the same sum formula was identified by Yao et al. [6] in a stoichiometric low-pressure *n*-heptane flame using MBMS. Their analysis of the photoionization threshold of $m/z = 114$ indicated that cyclic ethers may isomerize to open chain heptanones under flame conditions. Note again, in this study, we were not able to identify experimentally any specific structure of the respective species due to the chosen ionization method. However, the presence of doubly or singly oxygenated species close to the burner surface could be observed unambiguously. The experimental temperature profile in the first 2 mm (Fig. 1) shows temperatures between 500 and 1000 K which is in good agreement with the low-/intermediate-temperature regime of *n*-heptane.

Also the consumption of C_7H_{15} via the two other routes is involved. The different positions of either the radical site in C_7H_{15} or the double bond in C_7H_{14} give rise to a broad variety of decomposition pathways initiated by C–C-fissions and thus result in a complex mass flow pattern. This means that many intermediate C_2 , C_3 , C_4 , and C_5 species are formed by various pathways. This complexity is an effect of the fuel molecule's size and very different to the much simpler flame chemistry of butane, butene isomers and even 1-hexene, which were studied with older versions of the kinetic model [29,30,52]. Among the various decomposition pathways, two main threads of C-atom mass flows may be distinguished. Several consumption pathways of the C_7H_{15} and C_7H_{14} isomers lead via further decompositions to the formation of ethene or the preceding ethyl radical (see Fig. 3). Another important consumption pathway for C_7H_{15} leads, according to the mass flow analysis, to the formation of acetylene, either via a C_3 route (propene, allyl, allene, and propyne) ending with the decomposition of propyne or via a C_4 route starting with the C_4H_7 isomers and ending with the decomposition of $i-C_4H_3$ (see Supplement Material, Figs. S5 and S9). Ethene is converted to acetylene, which is then oxidized to form the ketylenyl radical (HCCO). The acetylene consumption to the stable products CO and CO_2 is very similar to other flames analyzed with the predecessor of the current kinetic model [29,30]. The rather complex picture of *n*-heptane consumption in flames is revealed by the C-atom mass flow analysis. The validity of this analysis is corroborated by the agreement between experimental and simulated species concentration profiles for many species, which will be discussed in the next section.

4.1.3. Major species

The consumption of *n*-heptane ($n-C_7H_{16}$) and oxygen (O_2) and the formation of the major products CO, CO_2 , H_2 , and H_2O is shown in Fig. 2. A very good agreement between the model prediction and experiment is observed for the fuel and the major products with the exception of H_2 . The deviation in the H_2 profiles can be explained with the calibration method for major species, since the concentrations of major radicals such as H and OH are not taken into account and their contribution is missing in the atom balance. The potential error of this assumption is strongest for H_2 . Taking the simulated concentration of 3.5% H-atoms into account, this effect would decrease the measured H_2 profile by

1.75% and a good agreement would then be achieved between model and experimental values. The O_2 profile close to the burner was found to be well predicted when using the calculated temperature profile as presented in Fig. 2 (see also discussion in Section 4.1.1).

4.1.4. Major intermediate species

As outlined in the discussion of *n*-heptane consumption above, a broad variety of intermediate C_2 to C_5 species are generated from the consumption of *n*-heptyl (C_7H_{15}) and *n*-heptene (C_7H_{14}) isomers. The chemical process is more complex than in the recently studied low-pressure flames of C_4 fuels [29,30]. The formation of C_2 to C_5 intermediates is specific for the studied *n*-heptane flame. We assume similarities in flames from larger molecules. Therefore, the simulation of this flame represents a critical test for the kinetic

model. The recent augmentations of the mechanism in order to simulate C_4 fuels were the necessary preconditions for the present study.

We first discuss the larger C_5 to C_7 intermediates. In the lower panels of Fig. 5 the summed-up experimental signal for C_7H_{14} is over-predicted in the simulation. However, the over-prediction was reduced from a factor of 16 obtained with the original version to a factor of 3 with the present model, including the revisions that were explained in the previous section. The key change was the application of kinetic data of Allara and Shaw [65] for the β -C–C fission of alkyl radicals. This reference was also cited for the chemical model presented by El Bakali et al. [3] in their flame study. Seemingly, detailed species measurement in *n*-heptane flames is a critical target experiment for the kinetic data regarding this reaction. The modeling of *n*-heptane combustion with a focus on

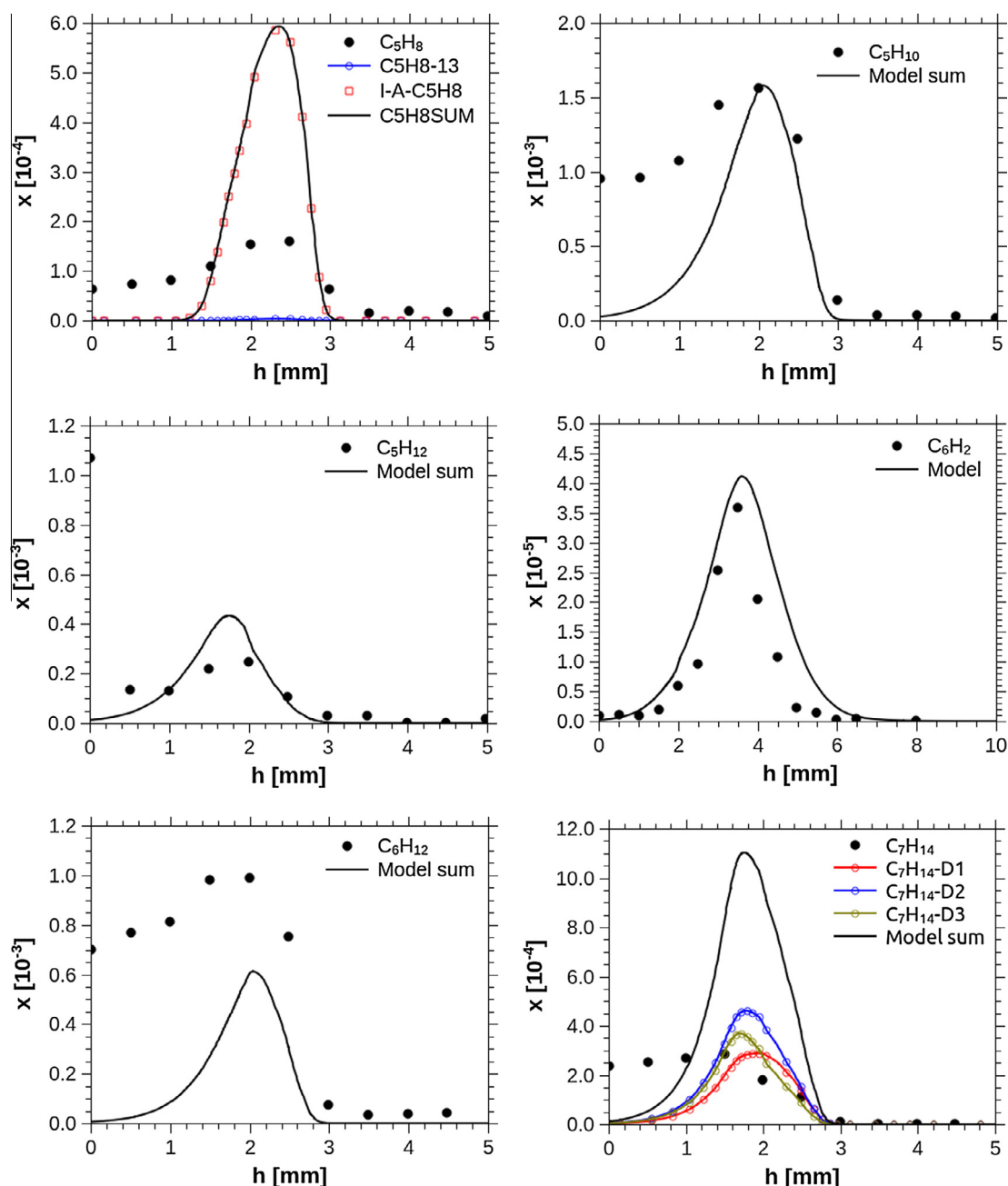


Fig. 5. Experimental values (filled symbols) for selected C_5 , C_6 , and C_7 species. Model predictions (lines and lines with open symbols) for C_5H_{10} , C_5H_{12} , and C_6H_{12} are only shown as sum of all isomers.

auto-ignition and speciation at lower temperatures may suggest the application of other kinetic data found in the literature. Figure 5 further illustrates that many C_5 and C_6 species (C_5H_{10} , C_5H_{12} , C_6H_{12} , C_6H_2) are well captured by the model while for C_5H_8 further model improvement will be possible. However, the isomeric composition of these intermediates is neither resolved in the experiment nor in the model. Therefore, a more detailed description of C_5 and C_6 chemistry could not be achieved in the framework of the present study. An intermediate step in this direction should be the extended study of detailed species compositions in C_5 and C_6 flames. Isomer-specific calibration of intermediates in such targets, including the fuel radicals and their degradation products, is desirable to provide the necessary chemical details for model development.

A very good agreement between experiment and model predictions is found for almost all C_4 intermediates. This is illustrated in Fig. 6. The agreement is probably a result of the detailed inspection of the flame chemistry of several C_4 flames in previous work [29,30]. The kinetic sub-model for C_4 species has not been changed in the present study. A very good agreement is also found for the stable C_2 species (compare Fig. 1). Predicted C_3 species concentrations are higher than the experimental data for C_3H_8 and C_3H_6 and agree well for the summed-up C_3H_4 isomers (see Fig. 7). Note that a close match is observed for the species profiles with highest concentrations and C-atom mass flows, ethene and acetylene.

The simulated profiles are also in good agreement for most of the measured radical species as shown in Fig. 8. The experimental

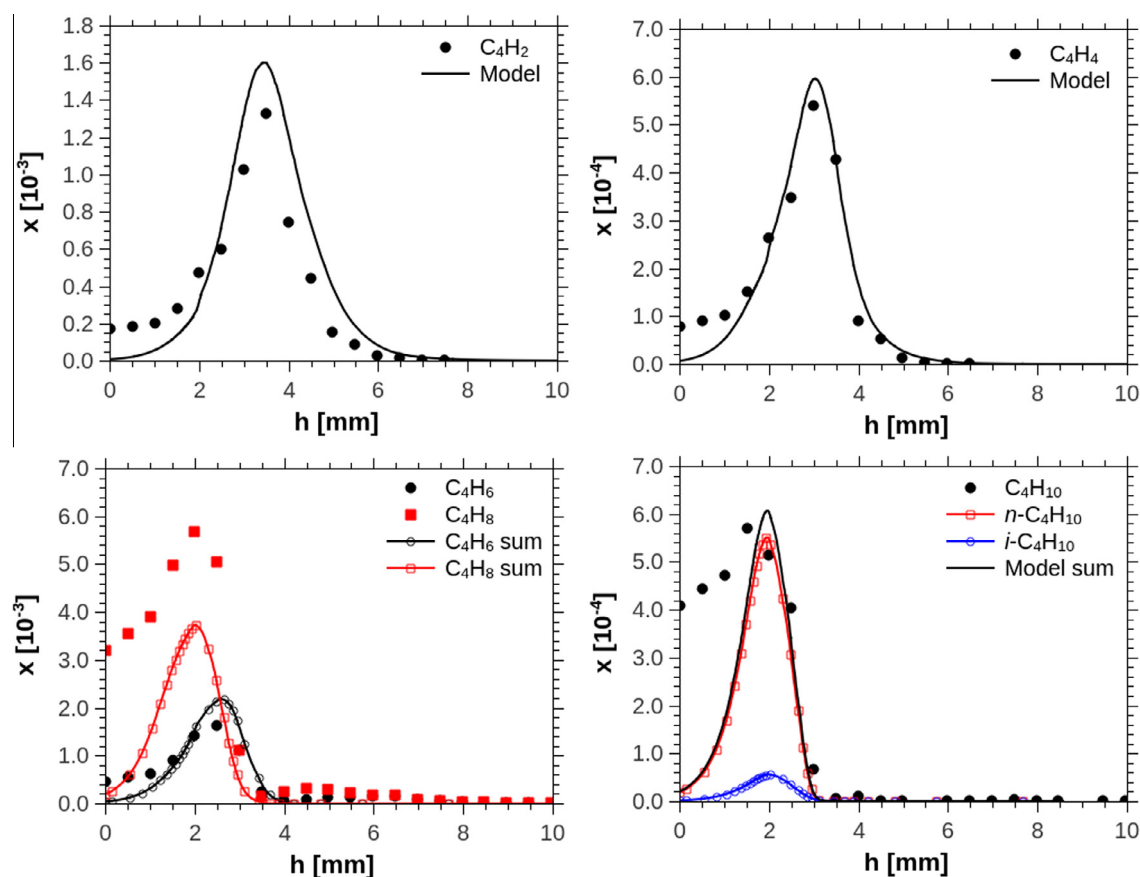


Fig. 6. Experimental values for C_4 species (filled symbols) and model predictions (lines and lines with open symbols). C_4H_6 and C_4H_8 are only shown as sum of all isomers.

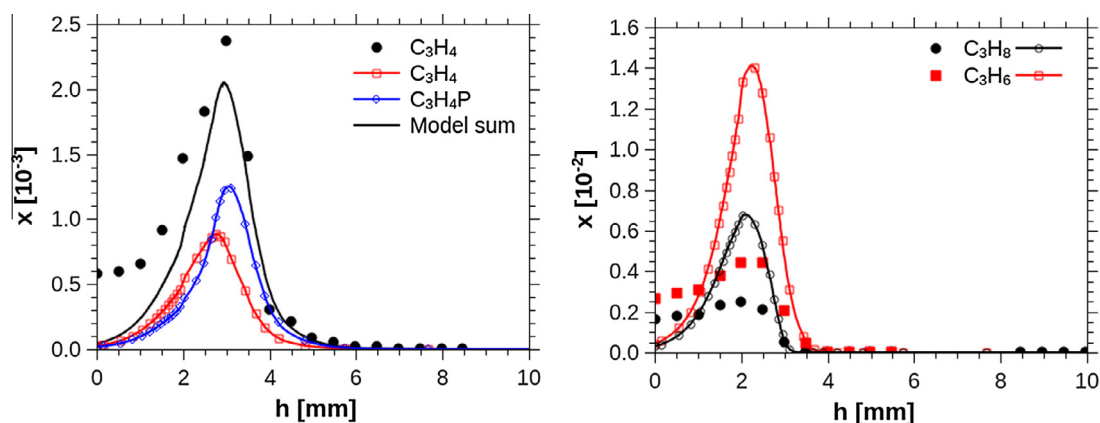


Fig. 7. Experimental values (filled symbols) and model predictions (lines and lines with open symbols) for C_3H_4 , C_3H_8 , and C_3H_6 .

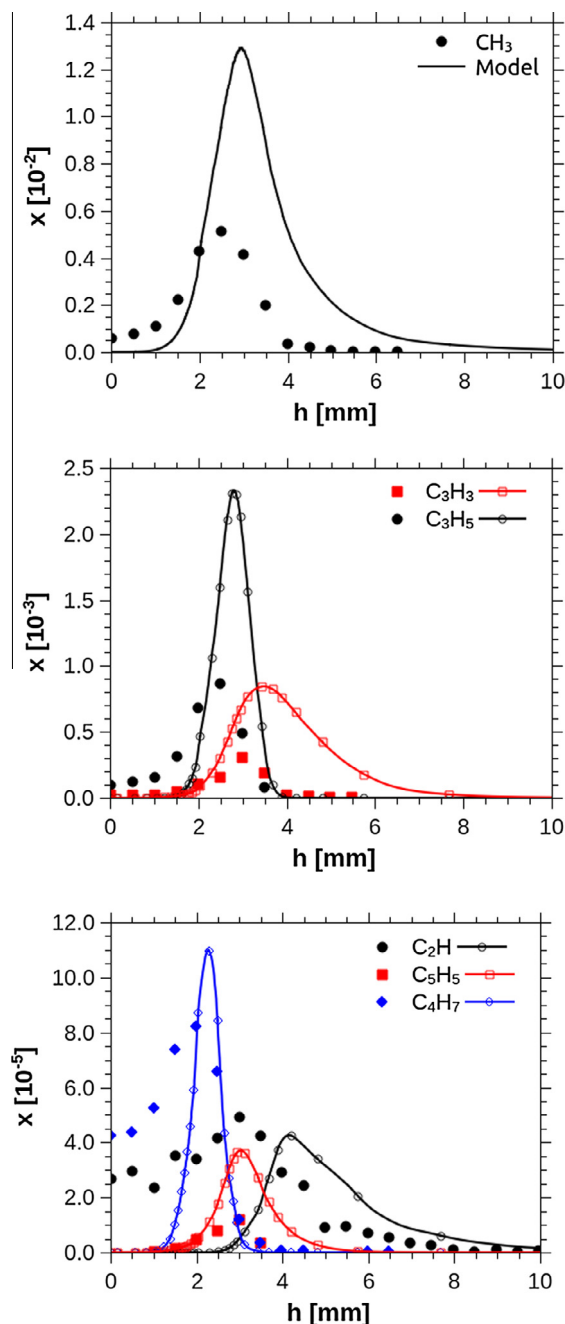


Fig. 8. Measurements (filled symbols) and model predictions (lines and lines with open symbols) for selected radicals: CH_3 , C_3H_3 , C_3H_5 , C_2H , C_4H_7 , and C_5H_5 . C_4H_7 is only shown as sum of all isomers.

uncertainty for radical mole fractions is significantly higher than for stable molecules. It should be noted that the concentrations of several radical species that are known to contribute to the formation of aromatic species (C_5H_5 , C_4H_5 , C_3H_5 , C_3H_3) are simulated within the experimental error range. The good overall agreement for these radicals and most major intermediates lends confidence to the analysis of the aromatic species formation pathways below.

4.1.5. Aromatic hydrocarbons

Several potentially aromatic species were detected in the experiment; C_6H_6 is assumed to be either benzene or fulvene; the detected species with the formula C_7H_8 is assumed to be toluene, and C_{10}H_8 is assumed to correspond to naphthalene. In Fig. 9 the

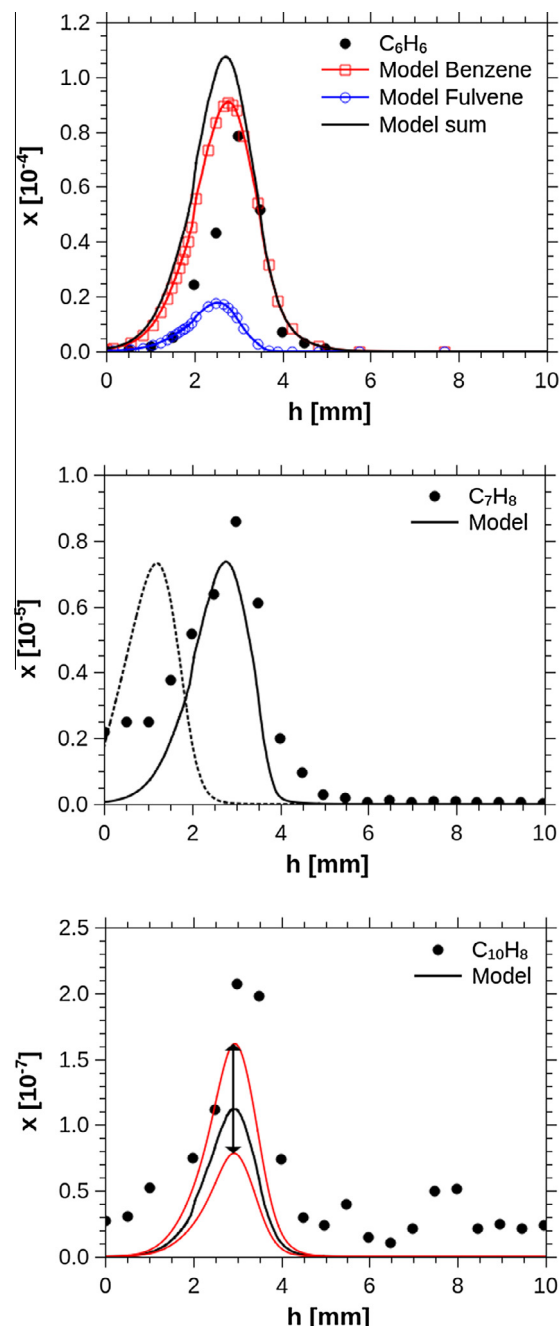


Fig. 9. Measured values (filled symbols) and model predictions (lines and lines with open symbols) for aromatic species: Benzene and fulvene (C_6H_6), toluene (C_7H_8), and naphthalene (C_{10}H_8). The red lines indicate the effect of changing the flame stoichiometry through uncertainty in the oxygen supply by $\pm 5\%$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

experimental mole fraction profiles are compared with the simulated ones for the named species. The simulations predict that the C_6H_6 signal belongs predominantly to benzene. It is slightly over-predicted but close to the assumed error margin of 30–40%. The experimental toluene profile is well captured by the simulation. The peak positions agree for benzene and toluene and also for the two-membered ring naphthalene, supporting the formation pathways assumed in the model. Note that C_{10}H_8 is observed close to the detection limit and the simulation also indicates a low mole fraction of this species. An additional comparison of the model with C_6H_6 and C_7H_8 profiles measured in C_4 flames, published in

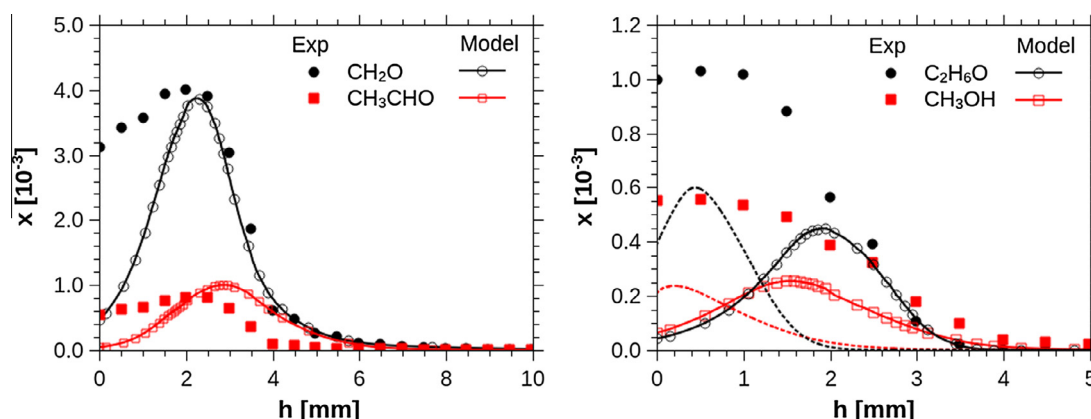


Fig. 10. Left: Measured values (symbols) for formaldehyde (CH_2O) and acetaldehyde (CH_3CHO) and model predictions (lines). Right: Methanol (CH_3OH) and $\text{C}_2\text{H}_6\text{O}$ (ethanol and dimethyl ether); measured values (symbols) and model predictions using the measured temperature profile (solid lines) and calculation with energy conservation (dotted lines).

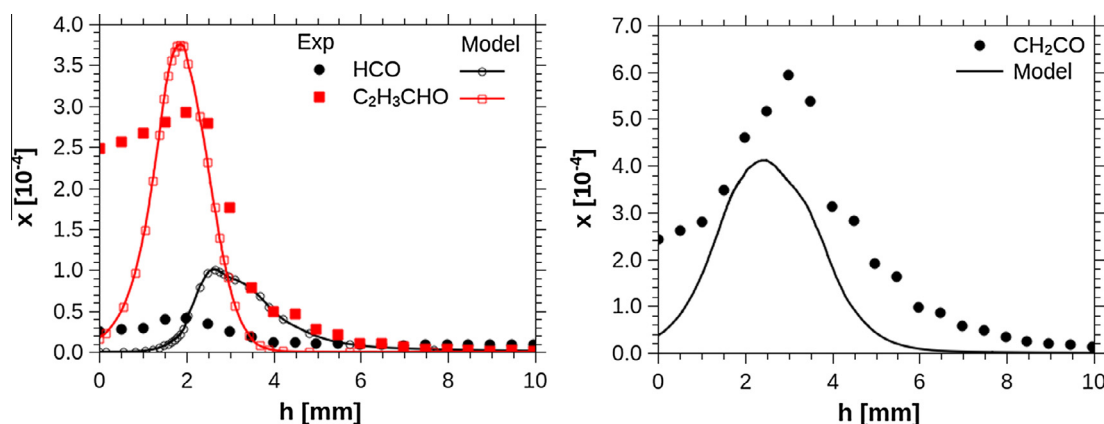


Fig. 11. Left: Measured values (symbols) for formyl radical (HCO) and propenal/acrolein ($\text{C}_2\text{H}_3\text{CHO}$) and model predictions (lines). Right: Ketene (CH_2CO) measurements and model predictions.

previous work can be found in the Supplement Material (Figs. S3 and S4).

The mass flow analysis has shown that more than 85% of the benzene is formed from the fulvene isomer. Fulvene in turn is formed via recombination reactions such as $\text{C}_5\text{H}_3 + \text{CH}_3$; $\text{C}_3\text{H}_3 + \text{C}_3\text{H}_5$ resulting in fulvene + 2H , $\text{C}_3\text{H}_3 + \text{C}_3\text{H}_3$, and several minor pathways. As outlined in Section 3, the fulvene pathways were included in the kinetic model and critically validated in very recent work on the flame chemistry of 1-hexene [52]. In that study fulvene and benzene were distinguished by their different ionization energies applying synchrotron-based single-photon ionization. The model captured both species for different stoichiometries. We note here that the older version of the kinetic model [18] could not reproduce C_6H_6 /benzene formation in *n*-heptane flames because of the missing fulvene chemistry. Toluene is formed from benzene via the phenyl radical in its subsequent reaction with methyl radicals.

The formation of C_{10}H_8 (naphthalene) is more complex. The dominant pathway under the present conditions (62%) is the combination of cyclopentadienyl radicals. The other pathways belong to the reaction of phenyl with C_4H_4 (21%) or the reaction of benzyl with propargyl radicals (11%). The remaining routes have low individual contributions (<6%). We note that a slightly higher ϕ of 1.78 at the upper limit of the error margin of 5% would significantly increase (ca. 20%) the predicted concentration of naphthalene and a lower ϕ would decrease it (see Fig. 9). Other aromatic species

are much less affected. Future experimental work should explore the formation of naphthalene and other polycyclic aromatic compounds under flame conditions slightly closer to the sooting point. Such studies might provide the critical reference data needed for augmenting the present comprehensive flame model to further explore the mechanisms that control the onset of soot formation [70].

4.1.6. Oxygenated intermediates

Regarding the formation of oxygenated intermediates, the first version of the model could not investigate this part of the *n*-heptane flame chemistry because of largely lacking experimental reference data [18]. The detected and simulated oxygenated species profiles are presented in Figs. 10 and 11. They include small aldehydes (CH_2O , CH_3CHO) and alcohols (CH_3OH , $\text{C}_2\text{H}_5\text{OH}$) as well as ketene ($\text{C}_2\text{H}_2\text{O}$) and acrolein ($\text{C}_2\text{H}_3\text{CHO}$). The experimental profiles of both aldehydes are well captured by the simulation (Fig. 10). About 67% of HCO is formed from CH_2O through numerous minor pathways from C_2 and C_3 species. Acetaldehyde (CH_3CHO) is formed via various pathways by C_2 to C_7 species and illustrates the complexity of the reaction scheme: about 11% are formed via the low-temperature pathway described in Section 4.1.2, in addition it is obtained from $\text{C}_6\text{H}_{12}\text{-D1}$ (4%), $\text{C}_5\text{H}_{10}\text{-D1}$ (9%), C_4H_6 (14%), C_3H_6 (19%), C_2H_5 (14%), and $\text{CH}_2\text{CH}_2\text{OH}$ (via C_2H_4 with 11%). It decomposes into a set of oxygenates: CH_3CO (38%), $\text{CH}_2\text{-CHO}$ (28%), CH_2CO (8%) and other products such as CH_4 (13%)

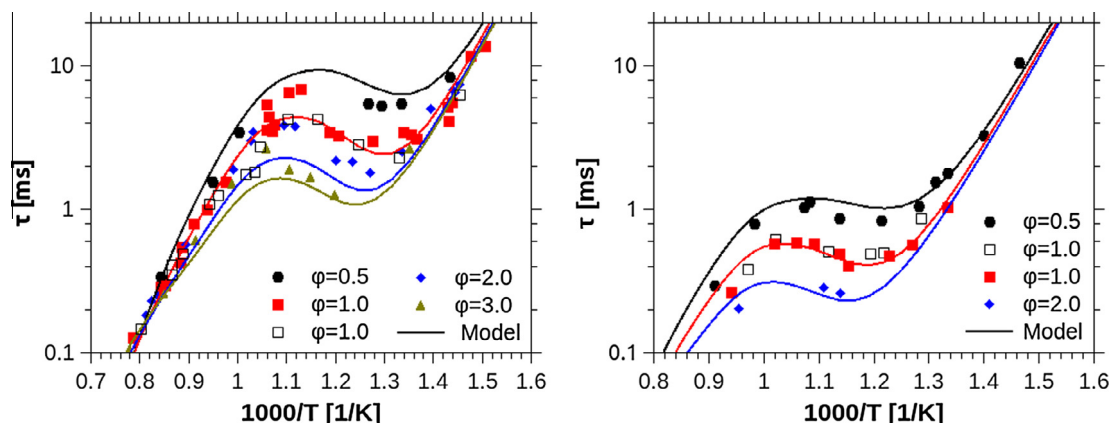


Fig. 12. Ignition delay times (τ) for *n*-heptane/air in a shock tube. *Left:* Experiments at 13.5 bar solid symbols [40]; open symbols [41]; solid lines: model predictions. *Right:* Ignition delay times for *n*-heptane/air in a shock tube at 40 ± 2 bar. Experiments: solid symbols for $\phi = 0.5$, $\phi = 1.0$ [39], and for $\phi = 2.0$ [40], open symbols [41]; solid lines correspond to model predictions at 40 bar.

and CO (10%). The formation of HCO is rather simple, it is formed via $\text{CH}_2\text{O} + \text{H}$ (65%) and reaction of CH with O_2 , CO_2 , and OH (18% in total); it decomposes into CO and H. The dehydrogenation of acetaldehyde is one path that produces ketene via the CH_2CHO radical. The mole fraction profile of ketene is reasonably well predicted as shown in Fig. 11.

The formation of the smallest alcohol methanol is slightly under-predicted by the kinetic model (see Fig. 10). Some caution is warranted here; the concentrations of these species peak close to the burner surface and this flame zone is problematic for comparison with the simulation as discussed in Section 4.1.1. According to the simulation methanol has three very different origins: Through the recombination of small radicals either $\text{CH}_3 + \text{OH}$ to form methanol directly (42%), via a formation route of the methylperoxy radical which subsequently reacts to methanol (31%) and directly from the fuel through H-abstraction via the methoxy radical (23%). $\text{C}_2\text{H}_6\text{O}$ is considered to be either ethanol or dimethyl ether. In the model $\text{C}_2\text{H}_6\text{O}$ is composed of 65% ethanol and of 35% dimethyl ether. In general $\text{C}_2\text{H}_6\text{O}$ is under-predicted by about a factor 2 while the maximum peak concentration differs between the prediction using the disturbed temperature profile and the undisturbed solution. Several larger detected oxygenated species such as propanol, butanol, and propionic acid are not included in the model. This gives room for future improvements. However, the oxygenated species with the highest mole fractions

(formaldehyde and acetaldehyde) are well captured by the model, and therefore an important part of the oxidation chemistry seems to be adequately represented.

4.2. Validation of the *n*-heptane model against other experimental targets

A short overview of the model validation against ignition delay time, flame speed, and jet-stirred reactor species measurements is provided in addition to highlight its wide applicability and support its comprehensive nature. As outlined above, the *n*-heptane oxidation sub-mechanism is largely unchanged compared to the original publication [18]. As expected, general trends of the model predictions have not changed and only the main results will be discussed here.

As an example, Fig. 12 compares the model predictions for the ignition delay times of undiluted *n*-heptane/air mixtures to those measured in shock tube experiments for temperatures between 600 K and 1200 K, fuel/air ratios between $\phi = 0.5$ –3.0 and pressures of 13 bar and 40 bar. The results of the earlier experiments [39,40] have been confirmed by Heufer and Olivier [41] in a recent study (see Fig. 12). The comparison shows a good agreement for the whole temperature and fuel oxygen mixture range. Especially the negative-temperature-coefficient (NTC) behavior is well captured by the simulation. The longest ignition delay times for the lean mixture

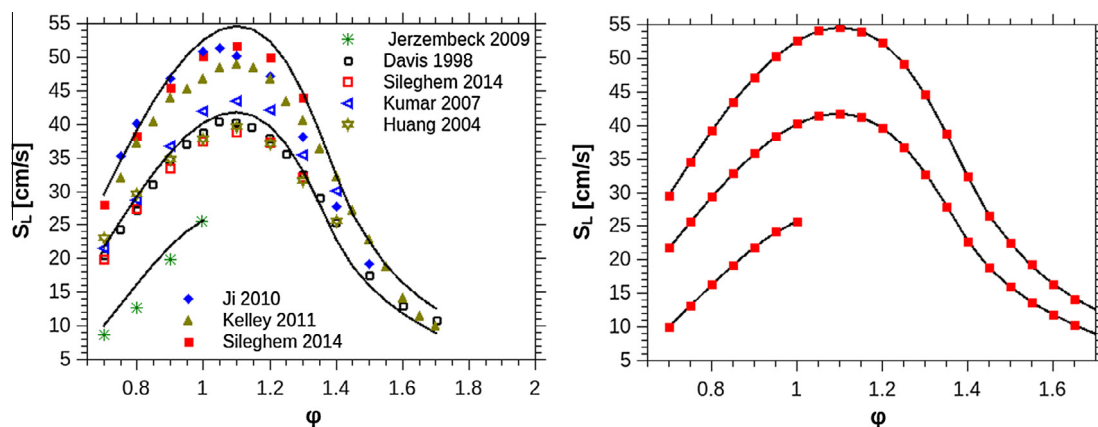


Fig. 13. *Left:* Laminar flame speed for *n*-heptane/air mixtures. Open symbols: Experiments at 298 K and 1 atm from [23,46,71]; Solid symbols: Experiments at 1 atm and 358 K [46], 353 K [47,73]; asterisks: 19.7 atm and 373 K [48]. Model prediction (solid lines) for 1 atm/298 K, 1 atm/358 K, and 19.7/373 K. *Right:* Predicted laminar flame speed with the complete reaction scheme (lines) and prediction obtained using a scheme without the low-temperature reactions for *n*-heptane (symbols).

seem to be slightly over-predicted. However, the deviation is within the experimental scatter. A further model comparison against shock tube data is shown in the Supplement Material (Fig. S1).

Figure 13 shows the comparison of laminar flame speeds for *n*-heptane as measured in several recent studies at cold-gas temperatures of 298 K, 353 K, and 373 K and pressures of 1 atm and 19.7 atm [23,46–48,71–73]. Good agreement is found for 298 K and 1 atm. The measured flame speeds in experiments at elevated cold-gas temperature are slightly over-predicted, while the prediction for elevated pressure and temperature exactly match the experimental data points. Figure 13 also shows that the laminar flame speed prediction is not influenced from the low-temperature chemistry since the inner layer temperature and adiabatic flame temperature of the freely propagating flame remain the same. A further model comparison against measured flame speeds for other C_1 – C_5 fuels is shown in the Supplement Material (Figs. S6 and S7).

An example of an analysis of the species composition in jet-stirred reactor experiments is shown in Figs. 14 and 15. Trends from work of Dagaut et al. [42] are reasonably well captured. The O_2 consumption in the cold flame region is somewhat under-predicted and hence CO formation is, too. The concentration profiles of the heptene isomers are reasonably well predicted but the distribution among the isomers differs by factors up to 2. Profile trends for acetylene and ethene are captured, but differences in the curvatures and concentrations are obvious even in the logarithmic plot. Clearly, a more detailed and improved validation of the kinetic model against jet-stirred reactor experiments including

recent studies [21] is desirable but beyond the scope of the present study with a focus on flame chemistry.

5. Summary and conclusion

The comparison of new experimental data concerning the species composition in a fuel-rich, low-pressure *n*-heptane flame with the predictions of the detailed kinetic model allowed a detailed inspection of the flame-specific *n*-heptane consumption routes. The mass flow analysis has revealed a complex flame chemistry with many parallel degradation pathways via C_2 to C_6 intermediates, including low-temperature oxidation reactions close to the burner surface that presumably form cyclic ethers or their isomerization products in detectable concentrations. The model predictions of soot precursors match the experimental profiles for most of the detected species. The peak positions and mole fractions of benzene and toluene are reasonably well predicted. The formation of naphthalene in low concentrations is consistently observed in experiment and simulation. The flow analysis revealed that benzene is predominantly produced from fulvene and not directly by propargyl radical combination as observed in previous work [29–31]. The much more detailed fulvene/benzene chemistry in the present model may thus affect the previous conclusions on benzene formation pathways. The main oxygenated products were found to be formaldehyde and acetaldehyde; their mole fractions are in close agreement with model predictions. The validity of the systematically-constructed kinetic model was corroborated

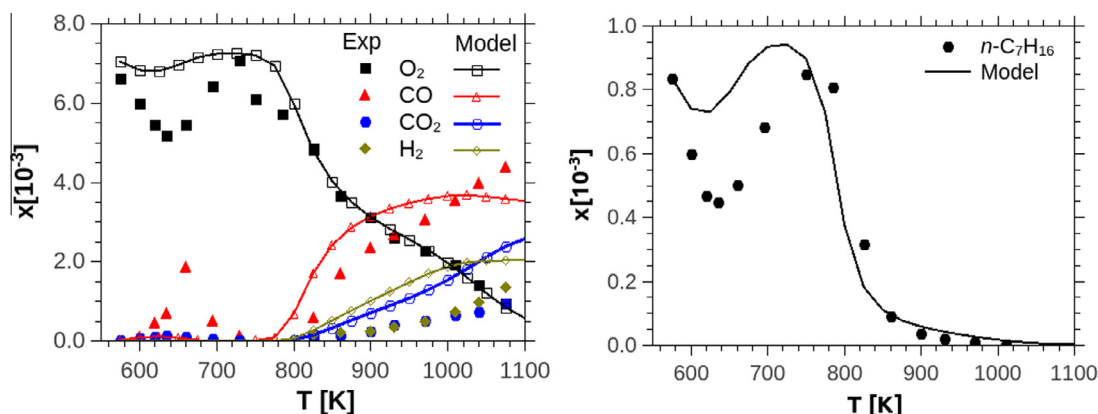


Fig. 14. Comparison of the model with measured species profiles (symbols) for major species in a jet-stirred reactor at 10 atm and $\varphi = 1.5$ from Dagaut et al. [42]; lines and lines with open symbols: model predictions.

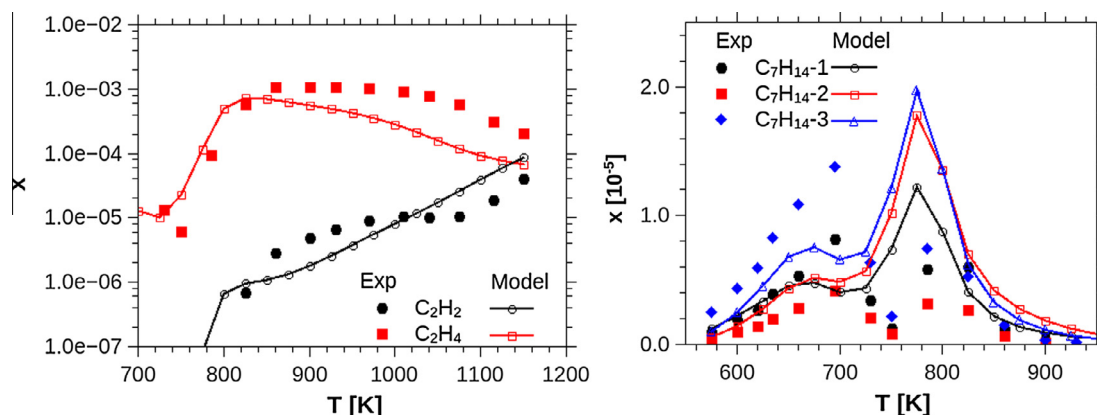


Fig. 15. Comparison of the model with measured species profiles for C_2H_2 and C_2H_4 (left) and C_7H_{14} isomers (right) at 10 atm and $\varphi = 1.5$ from Dagaut et al. [42]. The plotted experimental concentrations are the sum of the experimental *cis*- and *trans*-isomers for each position of the double bond.

by additional examination against other experiments including flame speeds, speciation in jet-stirred reactors and ignition delay time in shock tubes. Further detailed studies on isomer-specific species identification along the discussed pathways are recommended, including further quantitative information from low-temperature oxidation and combustion of C₅ and C₆ fuels which would support improvement of relevant sub-mechanisms.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2015.01.002>.

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